Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/18/2010
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Transgenic Plant Biomonitors of Space Flight Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/15/2007	**End Date:**	05/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS ; ISS NOTE: New end date is 5/31/2012, per PI (Ed., 2/3/2011) NOTE: Extended to 5/31/2011 per NSSC info; original end date 3/14/2010 (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Co-PI: Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX07AH27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is designed to be the next-generation of our previous flight project, PGIM-01, and investigates the use of reporter gene technology for the molecular analysis of development and stress in plants within space flight environments. We have designed a series of transgenic plants that carry sensor promoter - reporter gene constructs that are capable of monitoring a variety of environmental and developmental influences that might be altered in microgravity and general space flight environments. The first generation of these experiments with the GUS reporter have been tested for biological veracity on KC-135 parabolic flight missions, and flew as PGIM-01 in the Plant Growth Facility on STS-93. Results of these KC-135 missions and STS-93 were used in this proposal to (1) design a second generation of sensor promoters to refine our definition of the cellular biological impact of microgravity and space flight; (2) design a second generation of reporter gene constructs that include real-time and non- destructive observations of reporter gene activity; (3) design and conduct short and longer term flight experiments that begin a comprehensive analysis of molecular genetic changes induced by space flight; and (4) develop methods to integrate reporter gene activities with data from DNA chip analysis of genome expression. The long-term goal is an understanding of the space flight induced molecular changes in plant gene activity, an understanding that will lead to hardware and/or genetic engineering modifications that will mitigate the negative impacts of space flight on plant growth and development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of signal transduction in higher plants (and by inference, all eukaryotes) is central to basic molecular biology research.
Task Progress:	This year, the Payload Verification Test (PVT) was completed, along with several additional tests and experiments in preparation for the launch of the first component of the TAGES experiment (Run 1) on November 16th. The Imaging Hardware continued to undergo significant improvements right up to launch. In addition, insights gained during flight operations contributed to further improvements to the hardware and procedures. A brief summary of the TAGES flight experiment launched in 2009: Plants in space flight experience conditions that are distinctly unlike the earth-bound environments that have directed their evolution on Earth. This presents a unique opportunity to examine biological responses particularly those involved in integrating gravity as a force shaping biological systems. One of the consequences of exposure to any environment change is the expression of genes that allow survival in the new environment. In this case, understanding gene expression responses to spaceflight can illuminate the unique mechanisms employed in order to survive and adapt to an environment for which there is no evolutionary precedent. In this experiment gene expression will be monitored by genetically engineering plants with reporter genes driven by sensors that are sensitive to the potential stresses of spaceflight. These biological sensor plants are referred to as TAGES, an acronym for Transgenic Arabidopsis Gene Expression System. Those reporter genes are visualized as a green fluorescent glow that can be seen by imaging systems built within the spaceflight hardware carrying the experiment. The fluorescence is due to the Green Fluorescent Protein (GFP) portion of the reporter gene. The experiment will launch as part of the APEX-Cambium payload, a joint payload with the Canadian Space Agency. The payload will launch within the mid deck of Atlantis then will be transferred to an Express Rack on the ISS, where it will see multiple tests and experiments using the TAGES fluorescent plants. Samples will be returned on later shuttle missions but a significant component of the experimental data will be the fluorescent images that will be transmitted from the ISS on a regular basis over the next few months. Additional experiments will be launched on future shuttle missions to the ISS, carrying new plant types to be analyzed.In the fluorescent images captured by the GIS hardware, the fluorescent green color in the image indicates the location of cells responding to a particular environmental stress with the induction of the GFP reporter gene. The deep red color indicates the natural fluorescence of chlorophyll in healthy plant cells. Thus, green fluorescence is an indicator of a specific stress response, while the red fluorescence provides information on the general health and vigor of the plant. The GFP reporters can be engineered to respond to a variety of specific stresses. Using these types of reporters enables us to evaluate the biological impact of spaceflight thereby obtaining biological relevance of that environment, not just the physical values that may be measured by local instruments. The results from the first TAGES Flight experiment in 1999 (STS-93), and the results from several analog experiments with parabolic flight with (NASA's RGO) indicated that the spaceflight presents a complex environment to terrestrial organisms, an environment outside their evolutionary experience. We have used the insights gained from those previous data to design a set of experiments to help us understand further the biological impact of spaceflight on complex terrestrial organisms. Our in-flight fluorescent imaging capabilities allow us to evaluate the responses almost in real time with the GFP reporters, and our returned material will be subjected to powerful biochemical and genetic evaluations that will enable us to evaluate the global genetic and metabolic responses of plants; every gene can be analyzed. Plants are higher eukaryotic organisms, and as such, share a surprising amount (>40%) of genetic information and metabolic processes with other higher eukaryotes – like humans. The molecular information we obtain from plants can also be applied to analogous processes in humans.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Bamsey M, Berinstain A, Graham T, Neron P, Giroux R, Braham S, Ferl RJ, Paul A-L, Dixon MA. "Developing strategies for automated remote plant production systems: Environmental control and monitoring of the Arthur Clarke Mars Greenhouse in the Canadian High Arctic." Advances in Space Research 2009 Dec 15;44(17):1367-81. http://dx.doi.org/10.1016/j.asr.2009.08.012 , Dec-2009
Articles in Peer-reviewed Journals	Ferl RJ, Paul A-L. "Lunar Plant Biology: A Review the Apollo Era." Astrobiology, in press, January 2010. , Jan-2010
Articles in Peer-reviewed Journals	Visscher AM, Paul A-L, Kirst M, Alling AK, Silverstone S, Nechitailo G, Nelson M, Dempster WF, Van Thillo M, Allen JP, Ferl RJ. "Effects of a spaceflight environment on heritable changes in wheat gene expression." Astrobiology. 2009 May;9(4):359-67. PubMed PMID: 19413505 , May-2009

Significant Media Coverage	Ferl RJ, Paul A-L. "Sci-Path Mission to Mars: 'University of Florida plant scientists Rob Ferl and Anna-Lisa Paul talk about their efforts to understand how plants grow in space …' " Youtube: http://www.youtube.com/user/EXPLOREUF#p/u/1/UKKYNBXXCs4 , May 2009., May-2009